

Date: 03/10/2017, Tuesday Time:10.00 a.m to 1.00 p.m Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any TWO of the following:

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|---|--|----|
| A | a. Explain recycling of purine and pyrimidines by salvage pathway. | 5 |
| | b. Give explanation of sulphur cycle. | 5 |
| B | Explain de novo purine nucleotide synthesis from PRPP. | 10 |
| C | Add a short note on: | 10 |
| | a. DNA repair mechanisms | |
| | b. Mutagenesis | |

Q 2 Attempt any TWO of the following:

- | | | |
|---|--|----|
| A | Which enzymes are involved in DNA replication? Explain role of each enzymes and give account on significance of DNA replication. | 10 |
| B | a. Add a note on post translational modifications of proteins. | 5 |
| | b. Draw a pathway for biosynthesis of porphyrines. | 5 |
| C | Draw and explain Watson and Crick's model of DNA. | 10 |

SECTION- II

Q 3 Attempt any TWO of the following:

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|---|--|----|
| A | Add a detailed note on oxidative phosphorylation. | 10 |
| B | Discuss chemistry, absorption, metabolism and biochemical importance of vitamin A. | 10 |
| C | Explain electron transport chain in detail. | 10 |

Q 4 Attempt any TWO of the following:

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|---|---|----|
| A | Classify kidney function tests and explain any one in detail. | 10 |
| B | Add a note on: | 10 |
| | a. PCR | |
| | b. ELISA | |

C

Write a short note on:

a. Electrophoresis

b. Glucometer
